

NexteraAI Security Academy Module

Module 1: Introduction to Cybersecurity Protecting Your Digital Assets

Why Small Businesses in SA Need Cybersecurity Right Now

Why Your Business Can't Afford to Ignore Cybersecurity in 2026

South Africa is one of the most targeted countries in the world for cyberattacks—right behind the US and UK. Criminals don't just go after big corporations anymore. Small businesses like yours are prime targets because you often have valuable customer data, limited security, and less time to recover.

Shocking Stats (2025–2026):

- South African organisations face an average of **over 2,100–2,145 cyber attacks per week** (Check Point Software Technologies reports ~2,113 weekly in mid-2025, rising to 2,145 in early 2026 data—a 36% year-on-year jump in some metrics).
- This equates to roughly **577 attacks per hour** nationwide (consistent figure from Accenture and local reports still referenced in 2026).
- The average cost of a data breach in South Africa is now around **R44 million** (IBM Cost of a Data Breach Report 2025), down slightly from R53 million the year before thanks to better AI detection—but still devastating for small businesses.
- Many small businesses face **R43–50 million** in total damages when you include lost sales, recovery, fines, and reputation hits.
- Globally and locally, up to **60% of small businesses close within 6 months** of a major cyber incident (UK studies, but SA trends mirror this due to limited resources).

Your business holds customer details (ID numbers, phone numbers, card info, addresses). One breach can destroy trust, trigger POPIA fines (up to R10 million), and force closure. But the good news? Basic steps make a huge difference—and that's what this academy teaches.

The Three Main Goals (CIA Triad)

The CIA Triad – The Foundation of All Cybersecurity

Every strong defence aims to protect three core things. Think of them as the "CIA" of your business data:

1. **Confidentiality** – Keep customer data private Only authorised people see sensitive info. Example: Customer phone numbers, ID copies, or payment details stay hidden from hackers. A breach here leads to identity theft, fraud, or POPIA violations.
2. **Integrity** – Stop data being changed or deleted Ensure information stays accurate and untampered. Example: A hacker alters your stock records or changes a customer's booking—causing chaos, lost sales, or wrong charges.
3. **Availability** – Keep your systems working when you need them Your POS system, WhatsApp Business, online bookings, or CCTV must stay online. Example: Ransomware locks your files or a DDoS attack crashes your website—customers go elsewhere.

Protect all three, and your business stays safe, trusted, and running smoothly.



Everyday Examples for Retail, Salons, Mechanics, Restaurants

Real-Life Risks – How Cyberattacks Hit Businesses Like Yours

- **Retail Shop** — Hacker sends fake "supplier invoice" email (phishing). Owner clicks, malware steals card details from hundreds of customers. Result: Chargebacks, angry clients, POPIA report required.
- **Hair Salon/Beauty** — Staff use same weak password for booking app and email. Attacker guesses it, changes all appointments to blank, demands ransom. Salon loses a week's bookings (R20,000+ lost).
- **Mechanic/Auto Repair** — Customer pays via EFT, but criminal intercepts via fake "bank update" SMS. Mechanic's business account drained. No backups = weeks offline sorting it out.
- **Restaurant/Takeaway** — Online menu/order site hacked (weak security). Menu prices changed to R1, or orders deleted. Worse: Customer data leaked, leading to fraud and bad reviews.

These aren't "big company" problems. They happen weekly to SA small businesses—often starting with one click or missed software update.

Top 5 Threats Facing SA Small Businesses in 2026

The Top 5 Cyber Threats Hitting SA Small Businesses Right Now

1. **Phishing & Business Email Compromise (BEC)** — Fake emails/SMS/WhatsApp from "bank", "supplier", or "SARS". Most common entry point—tricks you into giving login details or clicking bad links.
2. **Ransomware** — Files locked, ransom demanded (often R100,000+). SMEs hit hard because many pay to recover quickly. AI makes these attacks smarter and faster in 2026.
3. **Weak Passwords & Credential Stuffing** — Reusing passwords across sites. Hackers use leaked lists to break in. Many small business accounts still use "123456" or "password".
4. **Missed Calls / Vishing / Smishing** — Criminals call pretending to be IT support, bank, or Microsoft. They trick you into remote access or sharing codes. Rising with load-shedding confusion.
5. **POPIA Risks & Data Leaks** — Accidental leaks (lost phone, unencrypted email) or breaches expose personal info. Fines up to R10 million + reputational damage. Regulator getting stricter in 2026.

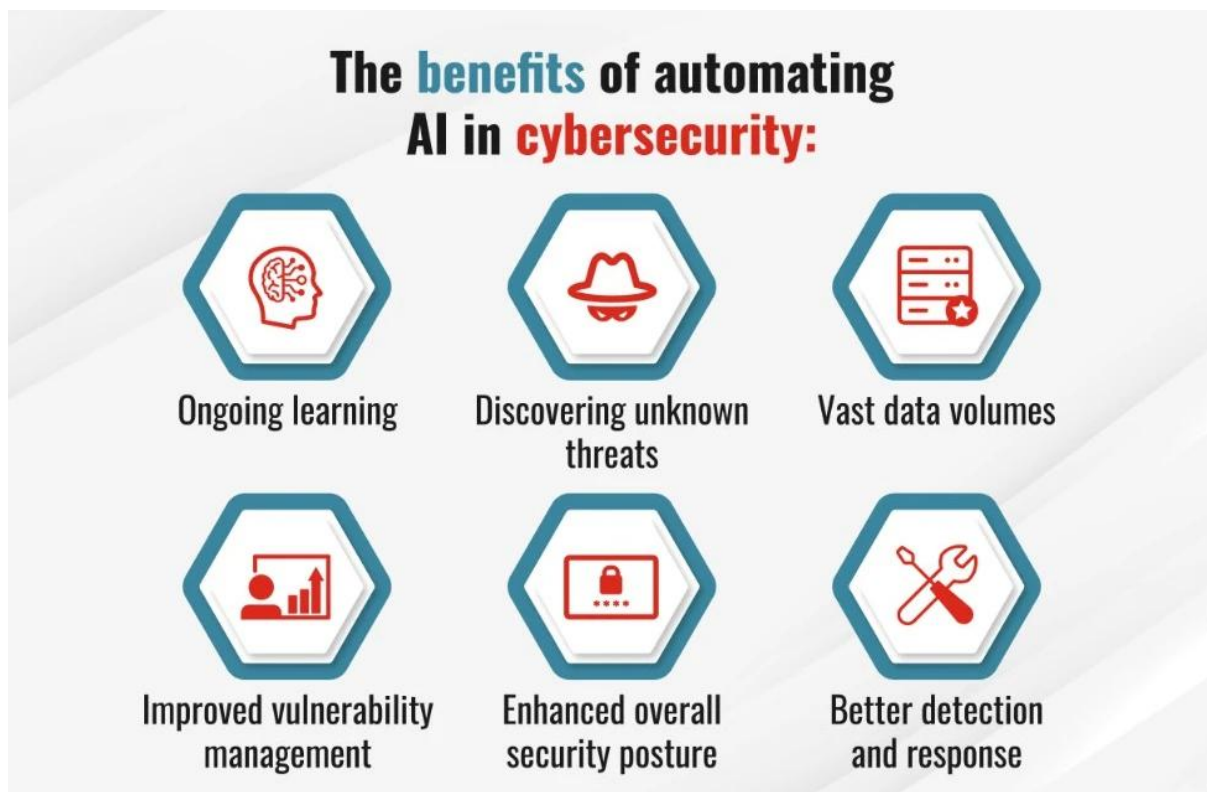
These threats exploit human habits and simple tech gaps—not high-tech wizardry.



5 Quick Actions Every Owner Can Take Today

5 Simple Steps You Can Implement This Week

1. **Turn on Multi-Factor Authentication (MFA)** everywhere — Email, banking, POS, WhatsApp Business. Use app or SMS code—not just password.
2. **Keep everything updated** — Phones, computers, POS software, routers. Auto-updates catch most vulnerabilities criminals exploit.
3. **Make regular backups** — Copy important files (invoices, customer lists, photos) to external drive **and** cloud (Google Drive/OneDrive with password). Test restoring once a month.
4. **Use a password manager** — Free ones like Bitwarden or LastPass generate strong, unique passwords. Stop reusing them.
5. **Train your staff** — 10-minute chat: "Don't click suspicious links", "Verify strange calls", "Report odd emails". Free resources available online.



What NexteraAI Security Does + Next Module Teaser + Quiz

How NexteraAI Security Helps SA Small Businesses Stay Safe

NexteraAI Security provides affordable, no-nonsense cybersecurity tailored for South African small businesses:

- Easy-to-use tools: MFA setup, password managers, backup automation.
- Monitoring & alerts for threats targeting SA (phishing, ransomware).
- POPIA compliance checklists and basic training.
- 24/7 support from local experts who understand load-shedding, EFT scams, and more.
- Scalable plans starting low—no big IT team needed.

We're here so you can focus on running your business, not fighting hackers.